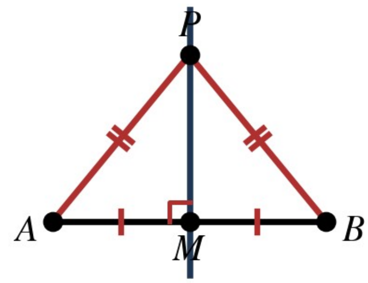
Speak **L**ike **A** **M**athematician:

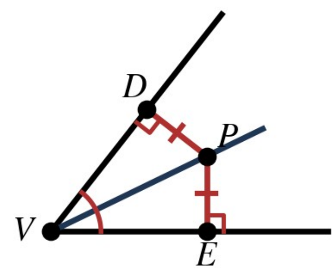
<u>equidistant:</u>

<u>perpendicular bisector:</u>

<u>angle bisector:</u>

<u>distance from a point to a line:</u>





Bisector Theorems

Perpendicular Bisector Theorem: if a point lies on the perpendicular bisector of a segment, then it is ____________________ from the ____________________ of the segment.

Converse: if a point is equidistant from the endpoints of a segment, then it lies on the ____________________ ____________________ of the segment.

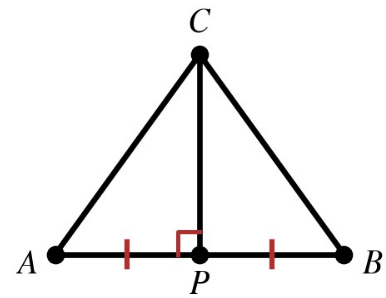
Angle Bisector Theorem: if a point lies on the bisector of an angle, then it is equidistant from the two ____________________ of the angle.

Converse: if a point in the interior of an angle is equidistant from the two sides, then it lies on the ____________________ of the angle.

Example 1 – Prove the Perpendicular Bisector Theorem

Given: $\overline{CP}$ is the perpendicular bisector of $\overline{AB}$

Prove: CA = CB

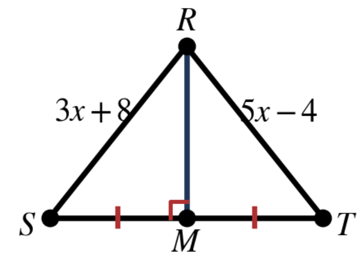


Statements	**Reasons**
1. $\overline{CP}$ is the perpendicular bisector of $\overline{AB}$	**1.** Given
2.	**2.**
3.	**3.**
4.	**4.**
5.	**5.**
6.	**6.**
7.	**7.**
8.	**8.**
9.	**9.**

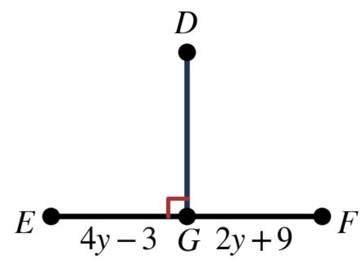
Example 2 – Find Each Measure

Use the Perpendicular Bisector Theorem and its converse.

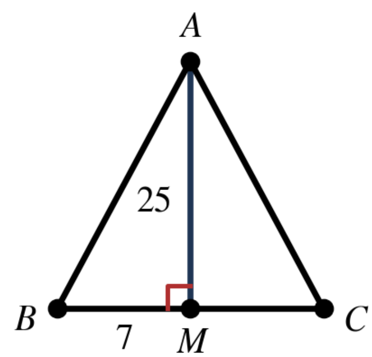
a) R lies on the perpendicular bisector of $\overline{ST}$. Find RS.



b) $\overline{DG}$ is the perpendicular bisector of $\overline{EF}$. Find EF.



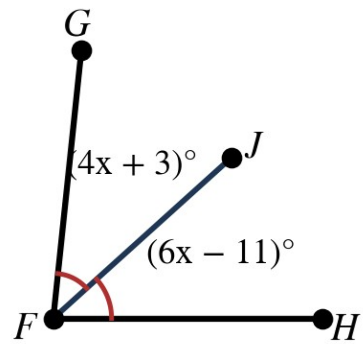
c) $\overline{AM}$ is the perpendicular bisector of $\overline{BC}$, AB = 25, and BM = 7. Find AM.



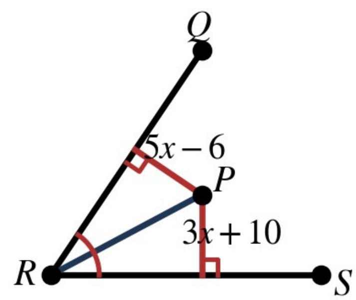
Example 3 – Angle Bisectors

Use the Angle Bisector Theorem and its converse.

a) $\overline{FJ}$ bisects ∠GFH. Find m∠GFJ.



b) P lies on the bisector of ∠QRS. Find PS.



c) $\overline{BD}$ bisects ∠ABC and m∠ABC = 74°. Find m∠ABD.

Example 4 – Write an Equation of a Perpendicular Bisector

Write an equation of the perpendicular bisector of the segment with endpoints P(−2, 3) and Q(4, 1).

a) Find the midpoint of $\overline{PQ}$.

b) Find the slope of $\overline{PQ}$, then the slope of the perpendicular bisector.

c) Write the equation. Then check that the midpoint satisfies it.

Example 5 – Use the Converses

Decide what each piece of information lets you conclude. Name the theorem.

a) Point K is 15 cm from R and 15 cm from S.

b) Point N is in the interior of ∠XYZ, and N is 6 in. from $\overline{YX}$ and 6 in. from $\overline{YZ}$.

c) A tent stake is driven so that it is the same distance from two poles. Where must it lie?